

Library Management System – Project Scenario

➤ Project Overview

As a product owner, I present the scenario for a Library Management System. This system involves designing a database to manage books, Users (Student, Faculty, External user), Staff members (Library Manager, Library Assistant, Library Technician, Librarian), and borrowing processes.

➤ Project Requirements

The requirements include creating database (LMS) using MySQL or XAMPP, designing entities (tables) for books , users, staff members, and borrowing processes using ER diagram, mapping ER diagram to a relational schema, identifying all Primary Keys (PK) and Foreign Keys (FK) in this schema and match every FK with its PK, creating a Data Dictionary that describes all entities, every relationship detail and every attribute definition, drawing the transaction pathway to illustrate the step-by-step data flow and logical sequence required to execute core queries and transactions within the database.

➤ Entity Analysis

1.The library has a number of books each book has book_id (Primary Key), book_name (Not Null), author_name (FK), release_year, total_copies (Default:1), available_copies,

book_description, book_cover, book_language (English || Arabic), and total_pages.

2.Many Users register in LMS each user has user_id (Primary Key), fullname, username (Unique), email (Unique), phone_number, user_type ('Student', 'Faculty', 'External_user'), max_borrow_limit, user_status ('Active', 'Suspended', 'Expired'), user_age (calculated field for age validation), password (must be hashed).

3.There are Staff Members who work in the library, each member has staff_id (Primary Key), staff_member_name (Not Null), email (Unique), phone_number, member_position (Library Manager, Library Assistant, Library Technician, Librarian), hire_date, salary, member_status ('Active', 'Inactive'), username (Unique), password (must be hashed), member_age (calculated field for age validation).

4.many borrowing processes done by users and every borrowing process has process_code (Primary Key), user_id (Foreign Key), book_id (Foreign Key), staff_id (Foreign Key), process_date, due_date, return_date, status ('Borrowed', 'Returned', 'Overdue').

5.every book has an author and every author has an author_name and country.

➤ Relationship Mapping & Cardinality

1.user (HAS) many borrowing processes

2. book (IS-Borrowed) many times
3. staff member (Makes) many borrowing processes
4. every author can (write) many books.

➤ Business Rules & Constraints

a) Age Validation Rules:

1. Minimum Age for Users Requirement: user must be at least 15 years old

2. Registration Block: System prevents registration if age < 15

4. Minimum Age for Staff Members Requirement: Members must be at least 25 years old

b) Data Integrity Rules:

1. available_copies cannot exceed total_copies

2. available_copies cannot be negative.

3. return_date must be $\geq$ process_date.

4. due_date must be $>$ process_date.

5. user cannot borrow if status is 'Suspended'.

6. Book cannot be borrowed if available_copies = 0.

➤ Operational Rules:

1. Borrowing Limits:

Student: 3 books

faculty: 10 books
external_user: 5 books

➤ Enhanced Sample Data Scenarios

Scenario 1: User Registration with Age Validation

1. New User applies with age = 13
2. **VALIDATION FAILED:** Age < 15 → Registration rejected
3. System displays: "Minimum age requirement is 15 years"

Scenario 2: Successful User Registration

1. New User applies with age = 20
2. **VALIDATION PASSED:** Age >= 15 → Registration approved
3. System displays: "Successful User Registration"

Scenario 3: Staff Member Registration with Age Validation

1. New member applies with age = 19
2. **VALIDATION FAILED:** Age < 25 → Registration rejected
3. System displays: "Minimum age requirement is 25 years"

Scenario 4: Successful Staff Member Registration

1. New User applies with age = 25
2. **VALIDATION PASSED:** Age >= 25 → Registration approved
3. System displays: "Successful Staff Member Registration"

GOOD Luck...